# Mining production report

## Weyland-Yutani Corporation

Sample snapshot from the supplied workbook

Period: 2099-11-02 to 2099-12-11. Generated 2026-11-03 17:26 UTC.

## Analysis settings

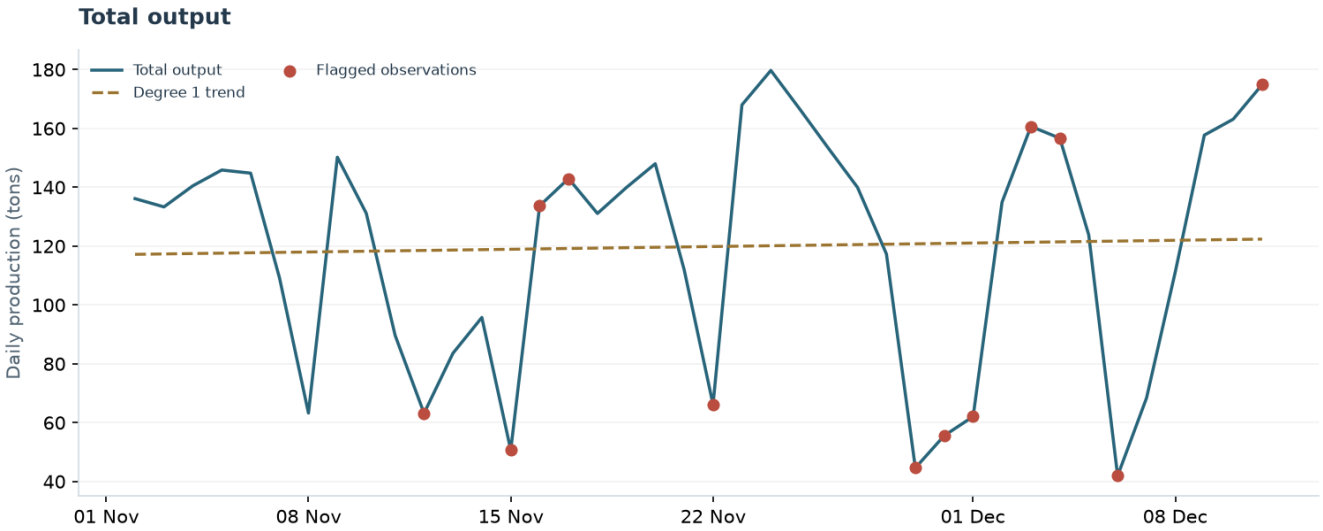
IQR multiplier 1.5; z-score threshold 3; moving-average window 7 previous observations; distance threshold 40%; Grubbs alpha 0.05. Charts: line, polynomial degree 1, highlight: Any test.

Standard deviation uses n-1. IQR uses linearly interpolated quartiles. The moving average excludes the current observation and needs a full window. Spike/drop direction compares output with that preceding mean, or the series median when the window is unavailable. Grubbs is a two-sided single-outlier test, at most one result per series. Its normality assumption can be violated by trends and weekly patterns; flags are candidates for review.

## Production statistics

| Series | Days | Mean | Std. dev. | Median | IQR |
|---|---|---|---|---|---|
| Total output | 40 | 119.78 | 40.34 | 133.51 | 60.42 |
| Fiorina 151 | 40 | 33.66 | 11.52 | 38.41 | 18.13 |
| LV-426 | 40 | 40.67 | 16.41 | 40.98 | 28.29 |
| Origae-6 | 40 | 45.45 | 16.31 | 49.09 | 22.43 |

# Total output




Flag counts - IQR: 0, Z-score: 0, Moving average: 12, Grubbs: 0. Distinct observations: 12.
IQR acceptance interval: -2.50 to 239.17 tons.
Grubbs: G = 1.9305; critical G = 3.0361.

## Anomaly details

### 2099-11-12: drop at 63.18 tons

Tests: Moving average. Reference output: 119.15 tons. Absolute z-score: 1.40. Distance from preceding moving average: 47.0%.

### 2099-11-15: drop at 50.67 tons

Tests: Moving average. Reference output: 96.68 tons. Absolute z-score: 1.71. Distance from preceding moving average: 47.6%.

### 2099-11-16: spike at 133.71 tons

Tests: Moving average. Reference output: 94.88 tons. Absolute z-score: 0.35. Distance from preceding moving average: 40.9%.

### 2099-11-17: spike at 142.95 tons

Tests: Moving average. Reference output: 92.53 tons. Absolute z-score: 0.57. Distance from preceding moving average: 54.5%.

### 2099-11-22: drop at 66.15 tons

Tests: Moving average. Reference output: 122.61 tons. Absolute z-score: 1.33. Distance from preceding moving average: 46.0%.

### 2099-11-29: drop at 44.61 tons

Tests: Moving average. Reference output: 141.54 tons. Absolute z-score: 1.86. Distance from preceding moving average: 68.5%.

### 2099-11-30: drop at 55.58 tons

Tests: Moving average. Reference output: 138.46 tons. Absolute z-score: 1.59. Distance from preceding moving average: 59.9%.

## 2099-12-01: drop at 62.07 tons

Tests: Moving average. Reference output: 122.41 tons. Absolute z-score: 1.43. Distance from preceding moving average: 49.3%.

## 2099-12-03: spike at 160.67 tons

Tests: Moving average. Reference output: 101.08 tons. Absolute z-score: 1.01. Distance from preceding moving average: 59.0%.

## 2099-12-04: spike at 156.66 tons

Tests: Moving average. Reference output: 102.16 tons. Absolute z-score: 0.91. Distance from preceding moving average: 53.4%.

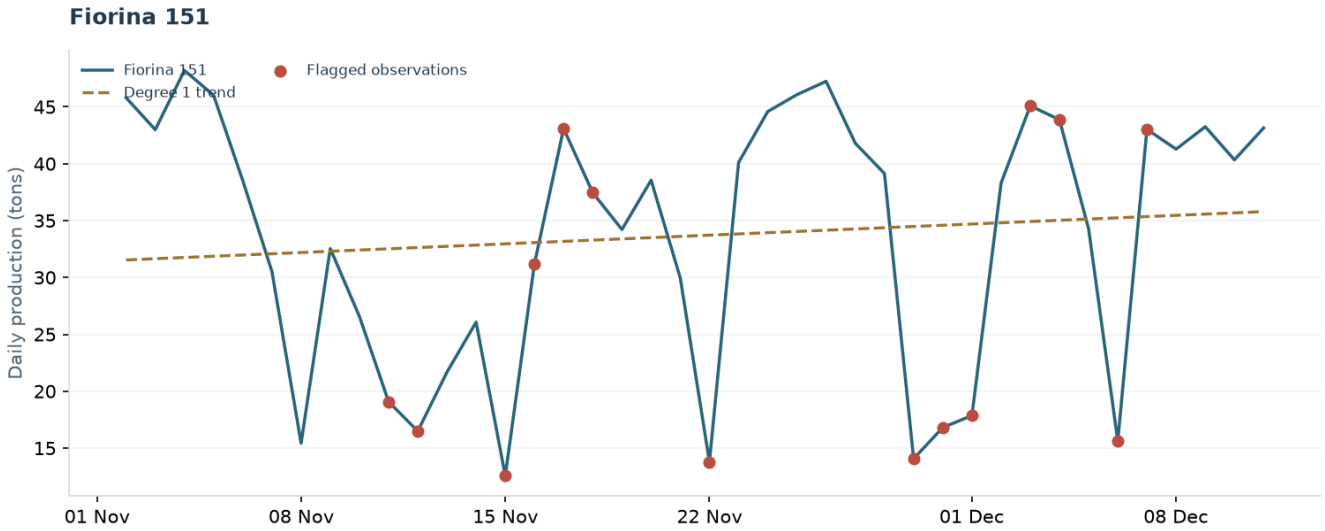
## 2099-12-06: drop at 41.91 tons

Tests: Moving average. Reference output: 105.47 tons. Absolute z-score: 1.93. Distance from preceding moving average: 60.3%.

## 2099-12-11: spike at 174.87 tons

Tests: Moving average. Reference output: 117.58 tons. Absolute z-score: 1.37. Distance from preceding moving average: 48.7%.

# Fiorina 151




Flag counts - IQR: 0, Z-score: 0, Moving average: 14, Grubbs: 0. Distinct observations: 14.
IQR acceptance interval: -2.23 to 70.30 tons.
Grubbs: G = 1.8312; critical G = 3.0361.

## Anomaly details

### 2099-11-11: drop at 19.08 tons

Tests: Moving average. Reference output: 33.97 tons. Absolute z-score: 1.27. Distance from preceding moving average: 43.8%.

### 2099-11-12: drop at 16.49 tons

Tests: Moving average. Reference output: 29.81 tons. Absolute z-score: 1.49. Distance from preceding moving average: 44.7%.

### 2099-11-15: drop at 12.56 tons

Tests: Moving average. Reference output: 22.55 tons. Absolute z-score: 1.83. Distance from preceding moving average: 44.3%.

### 2099-11-16: spike at 31.19 tons

Tests: Moving average. Reference output: 22.14 tons. Absolute z-score: 0.21. Distance from preceding moving average: 40.9%.

### 2099-11-17: spike at 43.09 tons

Tests: Moving average. Reference output: 21.95 tons. Absolute z-score: 0.82. Distance from preceding moving average: 96.3%.

### 2099-11-18: spike at 37.45 tons

Tests: Moving average. Reference output: 24.31 tons. Absolute z-score: 0.33. Distance from preceding moving average: 54.1%.

### 2099-11-22: drop at 13.79 tons

Tests: Moving average. Reference output: 32.43 tons. Absolute z-score: 1.72. Distance from preceding moving average: 57.5%.

### 2099-11-29: drop at 14.06 tons

Tests: Moving average. Reference output: 38.96 tons. Absolute z-score: 1.70. Distance from preceding moving average: 63.9%.

### 2099-11-30: drop at 16.81 tons

Tests: Moving average. Reference output: 39.00 tons. Absolute z-score: 1.46. Distance from preceding moving average: 56.9%.

### 2099-12-01: drop at 17.86 tons

Tests: Moving average. Reference output: 35.67 tons. Absolute z-score: 1.37. Distance from preceding moving average: 49.9%.

### 2099-12-03: spike at 45.09 tons

Tests: Moving average. Reference output: 30.74 tons. Absolute z-score: 0.99. Distance from preceding moving average: 46.7%.

### 2099-12-04: spike at 43.87 tons

Tests: Moving average. Reference output: 30.44 tons. Absolute z-score: 0.89. Distance from preceding moving average: 44.1%.

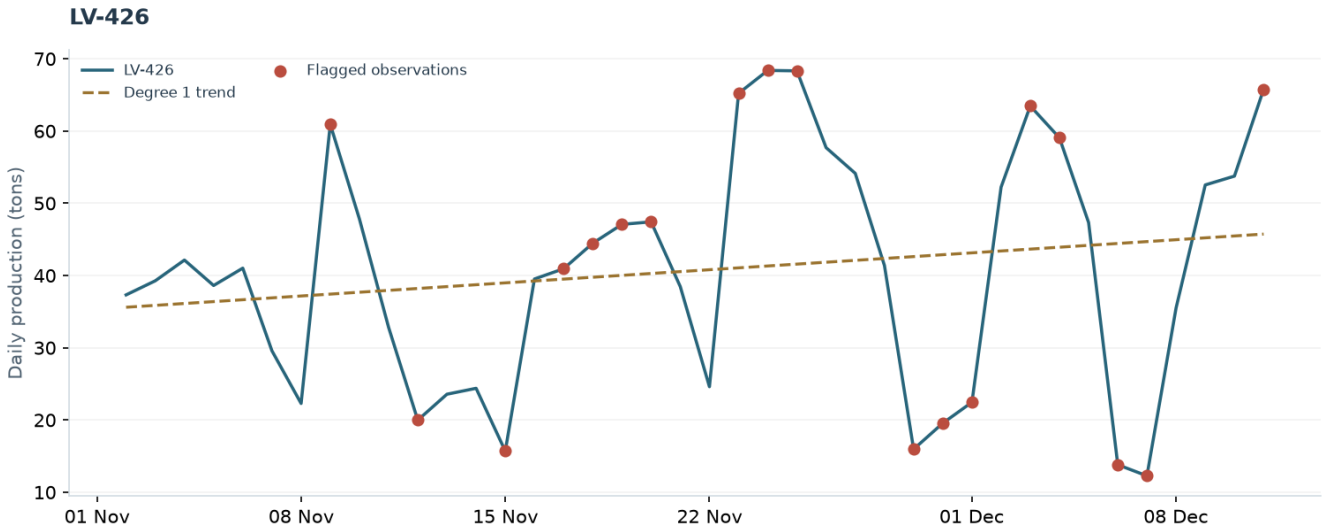
### 2099-12-06: drop at 15.61 tons

Tests: Moving average. Reference output: 30.04 tons. Absolute z-score: 1.57. Distance from preceding moving average: 48.0%.

### 2099-12-07: spike at 43.02 tons

Tests: Moving average. Reference output: 30.26 tons. Absolute z-score: 0.81. Distance from preceding moving average: 42.2%.

# LV-426




Flag counts - IQR: 0, Z-score: 0, Moving average: 18, Grubbs: 0. Distinct observations: 18.
IQR acceptance interval: -17.88 to 95.28 tons.
Grubbs: G = 1.7304; critical G = 3.0361.

## Anomaly details

### 2099-11-09: spike at 60.92 tons

Tests: Moving average. Reference output: 35.75 tons. Absolute z-score: 1.23. Distance from preceding moving average: 70.4%.

### 2099-11-12: drop at 19.98 tons

Tests: Moving average. Reference output: 39.01 tons. Absolute z-score: 1.26. Distance from preceding moving average: 48.8%.

### 2099-11-15: drop at 15.70 tons

Tests: Moving average. Reference output: 33.11 tons. Absolute z-score: 1.52. Distance from preceding moving average: 52.6%.

### 2099-11-17: spike at 40.94 tons

Tests: Moving average. Reference output: 29.11 tons. Absolute z-score: 0.02. Distance from preceding moving average: 40.6%.

### 2099-11-18: spike at 44.47 tons

Tests: Moving average. Reference output: 28.13 tons. Absolute z-score: 0.23. Distance from preceding moving average: 58.1%.

### 2099-11-19: spike at 47.08 tons

Tests: Moving average. Reference output: 29.80 tons. Absolute z-score: 0.39. Distance from preceding moving average: 58.0%.

### 2099-11-20: spike at 47.43 tons

Tests: Moving average. Reference output: 33.67 tons. Absolute z-score: 0.41. Distance from preceding moving average: 40.9%.

### 2099-11-23: spike at 65.22 tons

Tests: Moving average. Reference output: 40.36 tons. Absolute z-score: 1.50. Distance from preceding moving average: 61.6%.

### 2099-11-24: spike at 68.39 tons

Tests: Moving average. Reference output: 44.03 tons. Absolute z-score: 1.69. Distance from preceding moving average: 55.3%.

### 2099-11-25: spike at 68.33 tons

Tests: Moving average. Reference output: 47.95 tons. Absolute z-score: 1.69. Distance from preceding moving average: 42.5%.

### 2099-11-29: drop at 15.96 tons

Tests: Moving average. Reference output: 54.25 tons. Absolute z-score: 1.51. Distance from preceding moving average: 70.6%.

### 2099-11-30: drop at 19.56 tons

Tests: Moving average. Reference output: 53.02 tons. Absolute z-score: 1.29. Distance from preceding moving average: 63.1%.

### 2099-12-01: drop at 22.46 tons

Tests: Moving average. Reference output: 46.50 tons. Absolute z-score: 1.11. Distance from preceding moving average: 51.7%.

### 2099-12-03: spike at 63.47 tons

Tests: Moving average. Reference output: 37.64 tons. Absolute z-score: 1.39. Distance from preceding moving average: 68.6%.

### 2099-12-04: spike at 59.06 tons

Tests: Moving average. Reference output: 38.46 tons. Absolute z-score: 1.12. Distance from preceding moving average: 53.6%.

### 2099-12-06: drop at 13.81 tons

Tests: Moving average. Reference output: 40.01 tons. Absolute z-score: 1.64. Distance from preceding moving average: 65.5%.

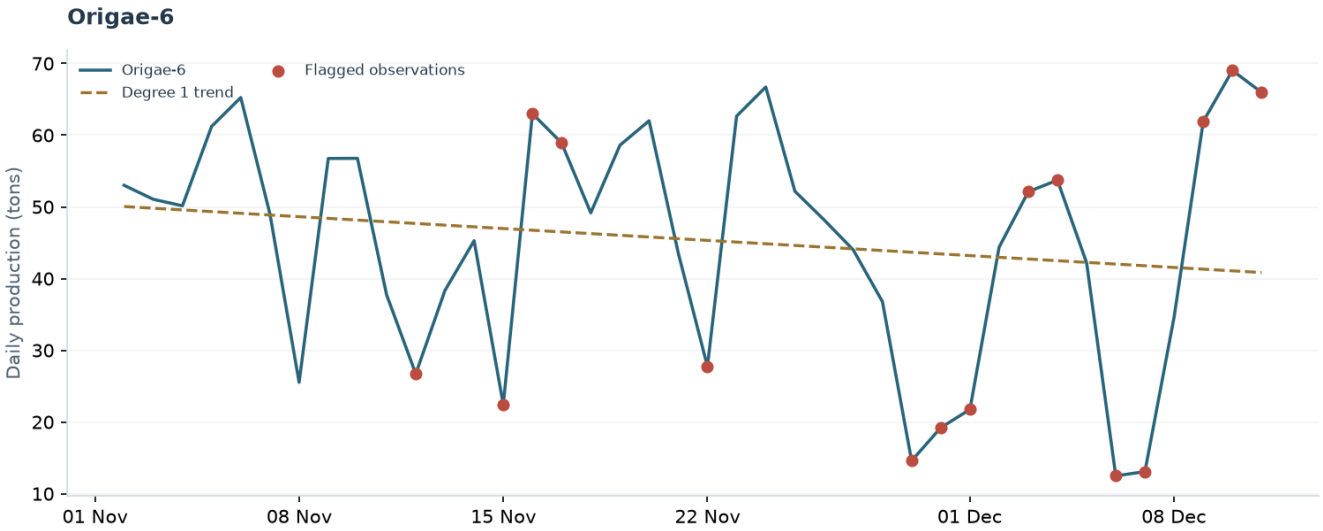
### 2099-12-07: drop at 12.27 tons

Tests: Moving average. Reference output: 39.70 tons. Absolute z-score: 1.73. Distance from preceding moving average: 69.1%.

### 2099-12-11: spike at 65.77 tons

Tests: Moving average. Reference output: 39.19 tons. Absolute z-score: 1.53. Distance from preceding moving average: 67.8%.

# Origae-6




Flag counts - IQR: 0, Z-score: 0, Moving average: 15, Grubbs: 0. Distinct observations: 15.
IQR acceptance interval: 2.59 to 92.33 tons.
Grubbs: G = 2.0200; critical G = 3.0361.

## Anomaly details

### 2099-11-12: drop at 26.71 tons

Tests: Moving average. Reference output: 50.33 tons. Absolute z-score: 1.15. Distance from preceding moving average: 46.9%.

### 2099-11-15: drop at 22.41 tons

Tests: Moving average. Reference output: 41.02 tons. Absolute z-score: 1.41. Distance from preceding moving average: 45.4%.

### 2099-11-16: spike at 63.00 tons

Tests: Moving average. Reference output: 40.57 tons. Absolute z-score: 1.08. Distance from preceding moving average: 55.3%.

### 2099-11-17: spike at 58.92 tons

Tests: Moving average. Reference output: 41.46 tons. Absolute z-score: 0.83. Distance from preceding moving average: 42.1%.

### 2099-11-22: drop at 27.75 tons

Tests: Moving average. Reference output: 51.10 tons. Absolute z-score: 1.08. Distance from preceding moving average: 45.7%.

### 2099-11-29: drop at 14.59 tons

Tests: Moving average. Reference output: 48.32 tons. Absolute z-score: 1.89. Distance from preceding moving average: 69.8%.

### 2099-11-30: drop at 19.21 tons

Tests: Moving average. Reference output: 46.44 tons. Absolute z-score: 1.61. Distance from preceding moving average: 58.6%.

### 2099-12-01: drop at 21.75 tons

Tests: Moving average. Reference output: 40.24 tons. Absolute z-score: 1.45. Distance from preceding moving average: 45.9%.

### 2099-12-03: spike at 52.11 tons

Tests: Moving average. Reference output: 32.70 tons. Absolute z-score: 0.41. Distance from preceding moving average: 59.4%.

### 2099-12-04: spike at 53.73 tons

Tests: Moving average. Reference output: 33.26 tons. Absolute z-score: 0.51. Distance from preceding moving average: 61.5%.

### 2099-12-06: drop at 12.49 tons

Tests: Moving average. Reference output: 35.42 tons. Absolute z-score: 2.02. Distance from preceding moving average: 64.7%.

### 2099-12-07: drop at 13.08 tons

Tests: Moving average. Reference output: 35.12 tons. Absolute z-score: 1.98. Distance from preceding moving average: 62.8%.

### 2099-12-09: spike at 61.92 tons

Tests: Moving average. Reference output: 36.09 tons. Absolute z-score: 1.01. Distance from preceding moving average: 71.6%.

### 2099-12-10: spike at 69.03 tons

Tests: Moving average. Reference output: 38.60 tons. Absolute z-score: 1.45. Distance from preceding moving average: 78.9%.

### 2099-12-11: spike at 65.96 tons

Tests: Moving average. Reference output: 41.01 tons. Absolute z-score: 1.26. Distance from preceding moving average: 60.8%.